

UNIT - I

1. Expand ASCII & EBCDIC (APRIL 2011)

ASCII: American Standard Code for Information Interchange.
EBCDIC: Extended Binary Coded Decimal Interchange Code

2. What is meant by positioned weighted number system? (APRIL 2011)

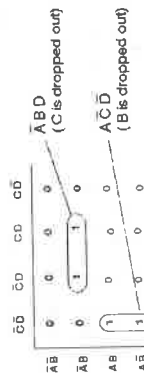
- > In a positional base- b numeral system (with b a natural number greater than 1 known as the radix), b basic symbols (or digits) corresponding to the first b natural numbers including zero are used.
- > To generate the rest of the numerals, the position of the symbol in the figure is used. For example, in the decimal system (base-10), the numeral 4327 means $(4 \times 10^3) + (3 \times 10^2) + (2 \times 10^1) + (7 \times 10^0)$, noting that $10^0 = 1$.
- > In general, if b is the base, one writes a number in the numeral system of base b by expressing it in the form $a_n b^n + a_{n-1} b^{n-1} + a_{n-2} b^{n-2} + \dots + a_1 b^1 + a_0 b^0$, and writing the enumerated digits $a_n a_{n-1} a_{n-2} \dots a_1 a_0$ in descending order. The digits are natural numbers between 0 and $b-1$.

3. Define Canonical form. (NOV 2011)

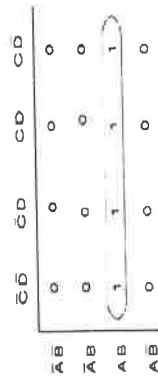
Any Boolean function can be expressed as a product of maxterms (with "product" meaning the ANDing of terms). The procedure for obtaining the product of maxterms directly from the truth table is as follows: Form a maxterm for each combination of the variables that produces a 0 in the function, and then form the AND of all those maxterms. **Boolean functions** expressed as a sum of minterms or product of maxterms are said to be in **canonical form**.

4. Define pairs and quads. (NOV 2011)

Pairs: Whenever you see a pair first encircle it and then simplify to get the simplified Boolean expression.



Quad: A group of 4 one's that are horizontally or vertically adjacent. End to end or in form of a square. A quad eliminates two variables and their complements.



5. State and prove De Morgan's theorem. (NOV 2012) (MAY 2014)

De Morgan's law 1: The complement of a product is equal to the sum of the complement.

$$\overline{AB} = \overline{A} + \overline{B}$$

A	B	A+B	(A+B)'	A'	B'	A'B'
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

De Morgan's law 2: The complement of a sum is equal to the product of the complement.

$$\overline{A+B} = \overline{A} \cdot \overline{B}$$

A	B	AB	(AB)'	A'	B'	A'B'
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

6. List the number systems?

- Decimal number systems
- Binary number system
- Octal number system
- Hexadecimal number system

7. Define the term digital? What is the best example of digital system?

The term digital refers to any process that is accomplished using discrete units.

Digital computer is the best example of digital system.

8. What is meant by map method?

The map method provides a simple straight forward procedure for minimizing Boolean function.

9. What is meant by two variable maps?

The variable map have four minterms for two variables, hence the map consists of four squares, one for each minterm.

10. What the use is of Don't care conditions?

Any digital circuit using this code operates under the assumption that these unused combinations will never occur as long as the system.

11. What is the algebraic function of Exclusive OR gate and Exclusive NOR gate?

$$F = xy' + x'y$$

$$F = xy + x'y'$$

12. What are the methods adopted to reduce Boolean function?

- Karnaugh Map
- Tabular method or Quine mc cluskey method.
- Boolean algebra & theorem.

13. Why we go in for tabulation method?

This method can be applied to problems with many variables and has the advantage of being suitable for machine computation.

14. What are prime implications?

The terms remained unchecked are called prime – implicants. They can't be reduced further.

15. Explain or list out the advantages and disadvantages of K- map methods?

The advantages of the k-map method are:

- It is a fast method for simplifying expression up to four variables.
- It gives a visual method of logic simplification.
- Prime implications and essential prime implicants are identified fast.
- Suitable for both SOP and POS forms of reduction.
- It is more suitable for class room teachings on logic simplification.

The disadvantages of the K – map method are:

- It is not suitable for computer reduction.
- K - Maps not suitable when the number of variables involved exceed four.
- Care must be taken to fill every cell with relevant entry, such as a 0, 1(or) don't care terms.
- Generally it is limited to six variable MAP. (i.e.) more than six variable involving expressions are not reduced.

16. What are the needs for binary codes?

- Code is used to represent letters, numbers and punctuation marks.
- Coding is required for maximum efficiency in single transmission.
- Binary code are the major components in the synthesis (artificial generation) of speech and video signals.
- By using error detecting codes, errors generated in signal transmission can be detected. Codes are used for data compression by which large amounts of data are transmitted in very short duration of time.

17. Mention the different type of binary codes?

The various types of binary codes are:

- BCD code (binary coded decimal).
- Self – complementing code.
- The excess -3(X's-3).
- Gray code.
- Binary weighted code.
- Alphanumeric code.
- The ASCII and EBCDIC code.
- Error – detecting and error – correcting code.
- Hamming code.

18. What is the truth table?

A truth table lists all possible combinations of inputs and the corresponding outputs.

19. Name the two basic types of Boolean expression?

There are two basic types of Boolean expression.

- Sum of product (SOP)
- Product of sum (POS)

20. Define Gray code.

The term "Gray code" is sometimes used to refer to any single-distance code, that is, one in which adjacent code words (perhaps representing integers differing by 1) differ by 1 in one digit position only.

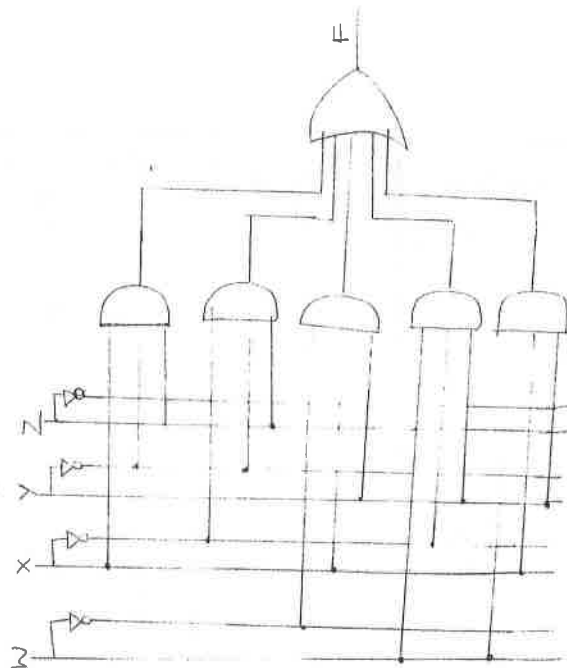
21. Define excess - 3 code.

A code which represents numbers as the four-bit binary equivalent of a decimal digit plus three. For instance, the number 0 is represented by 0011, which is 3 (decimal 0 + 3). Its abbreviation is excess 3 code.

18) Given the following Boolean function. (Nov'11)

$$F = xy'z + x'y'z + w'xy + wx'y + wxy$$

(a) Draw the Logic diagram for F. (6 marks)



(b) Simplify the function to a minimum no. of literals using Boolean Algebra. (5 marks)

$$\begin{aligned} F &= xy'z + x'y'z + w'xy + wx'y + wxy \\ &= y'z(x+x') + xy(w+w') + wx'y \\ &= y'z + xy + wx'y \end{aligned}$$

(20)

19) Determine the Prime-implicants of the function: (11) (APR/MAY'12)

$$F(w, x, y, z) = \sum (1, 4, 6, 7, 8, 9, 10, 11, 15)$$

$$\begin{aligned} F(w, x, y, z) &= (1, 9) + (4, 6) + (6, 7) + (7, 15) + (8, 9, 10, 11) \\ &= (01-0) + (01-0) + (01-1) + (-1-1) + (10-1) \\ &= x'yz + wxz + wxy + xyz + wx \end{aligned}$$

Step 4: Table of Essential Prime Implicants

Prime Implicants	1	4	6	7	8	9	10	11	15
1, 9	✓								
4, 6		✓							
6, 7			✓						
7, 15				✓					
8, 9, 10, 11					✓	✓	✓	✓	

First Report

Step 2	Step 3	Step 4	Step 5
1, 9	0001	✓	1, 9
4, 6	0100	✓	4, 6
8, 9	1000	✓	8, 9
8, 10	10-0	✓	8, 10
6, 7	011-	✓	6, 7
9, 11	10-1	✓	9, 11
10, 11	101-	✓	10, 11
10, 15	1111	✓	10, 15
11, 15	1-11	✓	11, 15

Second Report

Step 3

Step 4

Step 5

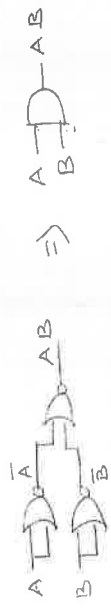
18

NOR as a Universal Gate

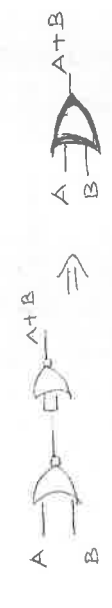
Implementation of NOT using NOR



Implementation of AND using NOR



Implementation of OR using NOR



Implementation of OR using NAND



19

Draw the Logic circuit for the

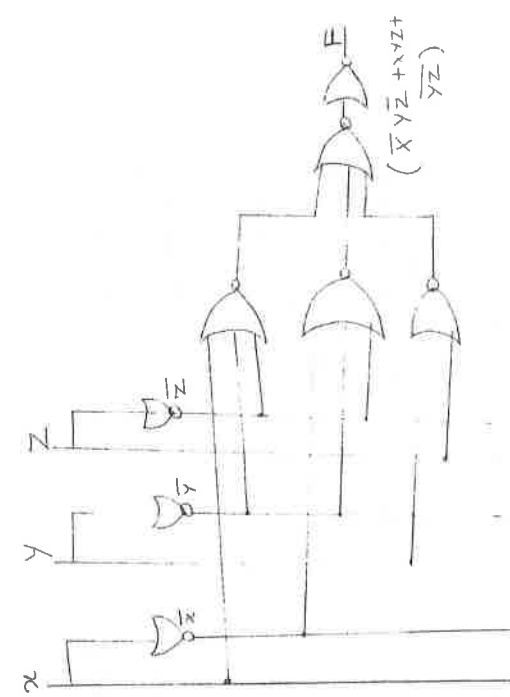
Boolean $\overline{X}Y\overline{Z} + XYZ + \overline{Y}Z$ function, using NOR gates. (NOV'08)

$$F = \overline{X}Y\overline{Z} + XYZ + \overline{Y}Z$$

$$\overline{F} = (\overline{X}Y\overline{Z} + XYZ + \overline{Y}Z) \quad (\because \overline{\overline{X}} = X)$$

$$= (\overline{X+Y+Z}) \quad (\text{OR}) \quad (Y+Z)$$

$$= (\overline{X+Y+Z})' \quad (\overline{X'+Y'+Z'})' \quad (Y+Z)'$$



①

DIGITAL SYSTEM DESIGN - 12/53

UNIT-I

11 MARK

1. Convert the following numbers to the indicated bases.
(NOV'12)
- (a) Decimal 225 to binary, Octal and Hexadecimal.
- (b) Binary (11010111) to decimal, Octal and Hexadecimal

$$(a) (225)_{10} \text{ to } ()_2$$

2	2	2	5
2	1	1	2
2	5	6	- 1 (a ⁰)
2	1	4	- 0 (a ¹)
2	2	8	- 0 (a ²)
2	1	4	- 0 (a ³)
2	7	- 0 (a ⁴)	
2	3	- 1 (a ⁵)	
	1	- 1 (a ⁶)	

$$\Rightarrow (225)_{10} = (a^7 a^6 a^5 a^4 a^3 a^2 a^1 a^0) = (11100001)_2 //$$

$$(b) (225)_{10} \text{ to } ()_8$$

8	2	2	5
8	2	8	- 1 (a ⁰)
	3	- 4 (a ¹)	

$$\Rightarrow (225)_{10} = (a^2 a^1 a^0) = (341)_8 //$$

$$(c) (225)_{10} \text{ to } ()_{16}$$

16	2	2	5
	1	4	- 1 (a ⁰)

$$\Rightarrow (225)_{10} = (a^1 a^0) = (141)_{16} = (E1)_{16}$$

$$(d) (11010111)_2 \text{ to } ()_{10}$$

$$\Rightarrow (11010111)_2$$

$$\Rightarrow (1 \times 2^7) + (1 \times 2^6) + (0 \times 2^5) + (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (1 \times 2^1) + (1 \times 2^0)$$

$$\Rightarrow 128 + 64 + 0 + 16 + 0 + 4 + 2 + 1$$

$$\Rightarrow (215)_{10}$$

$$\Rightarrow (11010111)_2 = (215)_{10}$$

$$(f) (11010111)_2 \text{ to } ()_{16}$$

$$\Rightarrow (11010111)_2$$

$$\Rightarrow (011010111)_2$$

$$\Rightarrow (327)_8$$

$$\Rightarrow (11010111)_2 = (327)_8$$

$$\Rightarrow (11010111)_2 = (D7)_{16}$$

2. Give the decimal equivalent of the following binary numbers.

$$(a) (11010111)_2 \text{ to } ()_{10}$$

$$\Rightarrow (11010111)_2$$

$$\Rightarrow [(1 \times 2^7) + (1 \times 2^6) + (0 \times 2^5) + (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (1 \times 2^1) + (1 \times 2^0)]$$

$$\Rightarrow 32 + 16 + 0 + 4 + 0 + 1 + 1 = 53$$

$$\Rightarrow (11010111)_2 = (53)_{10}$$

(APR'04)

③

$$\begin{aligned}
 (b) \quad & (11001.011)_2 \text{ to } ()_{10} \\
 \Rightarrow & (1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 + 0 \cdot 2^{-1} + 1 \cdot 2^{-2} + 1 \cdot 2^{-3})_{10} \\
 \Rightarrow & [(1 \times 2^4) + (1 \times 2^3) + (0 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)] + [(0 \times 2^{-1}) + (1 \times 2^{-2}) + (1 \times 2^{-3})] \\
 \Rightarrow & [16 + 8 + 0 + 0 + 1] + [0 + \frac{1}{4} + \frac{1}{8}] \\
 \Rightarrow & [25] + [0.25 + 0.125] \\
 \Rightarrow & (25.375)_{10} \\
 \Rightarrow & (11001.011)_2 = (25.375)_{10}
 \end{aligned}$$

2. (b) Convert the following Hexadecimal numbers to binary (APR 04)

(i) E5 (ii) B4D (iii) 74F4

$$\begin{aligned}
 (i) \quad & (E5)_{16} \text{ to } ()_2 \\
 \Rightarrow & (E5)_{16} = (1110 \ 0101)_2
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad & (B4D)_{16} \text{ to } ()_2 \\
 \Rightarrow & (B4D)_{16} = (1011 \ 0100 \ 1101)_2
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad & (74F4)_{16} \text{ to } ()_2 \\
 \Rightarrow & (74F4)_{16} = (0111 \ 0100 \ 1111 \ 0100)_2
 \end{aligned}$$

④

3. Convert the following numbers into decimal number system using their positional weight (4 marks) (APR 11)

(i) $(1A7)_{16}$ (ii) $(1427)_8$

$$\begin{aligned}
 (i) \quad & (1A7)_{16} \text{ to } ()_{10} \\
 \Rightarrow & [(1 \times 16^2) + (10 \times 16^1) + (7 \times 16^0)] \\
 \Rightarrow & 256 + 160 + 7 \\
 \Rightarrow & (423)_{10} \\
 \Rightarrow & (1A7)_{16} = (423)_{10}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad & (1427)_8 \text{ to } ()_{10} \\
 \Rightarrow & [(1 \times 8^3) + (4 \times 8^2) + (2 \times 8^1) + (7 \times 8^0)] \\
 \Rightarrow & (512 + 256 + 16 + 7) \\
 \Rightarrow & (791)_{10} \\
 \Rightarrow & (1427)_8 = (791)_{10}
 \end{aligned}$$

4) Convert the decimal 250.5 to base 2, base 4, base 8 and base 16.

$$\begin{aligned}
 (a) \quad & (250.5)_{10} \text{ to } ()_2 \\
 \Rightarrow & (250.5)_{10} = (a^3 a^2 a^1 a^0 . a^{-1})_2 \\
 = & (11111010.1)_2
 \end{aligned}$$

$$\begin{array}{r}
 2 \overline{) 250} \\
 \underline{2} \\
 0 \\
 2 \overline{) 25} \\
 \underline{2} \\
 0 \\
 2 \overline{) 5} \\
 \underline{4} \\
 1 \\
 2 \overline{) 1} \\
 \underline{2} \\
 0
 \end{array}$$

$$0.5 \times 2 = 1.0 \quad (a^{-1})$$

(5)

(b) $(250.5)_{10}$ to $(\)_4$

$\downarrow$
 $(250) + (0.5)$

$\downarrow$
 $0.5 \times 4 = 2.0(a^1)$

$4 \overline{) 250}$
 $4 \overline{) 62} - 2(a^2)$
 $4 \overline{) 15} - 3(a^1)$
 $3 - 3(a^0)$

$\Rightarrow (250.5)_{10} = (a^2 a^1 a^0 a^{-1})_4 = (3322.2)_4$

(c) $(250.5)_{10}$ to $(\)_8$

$\downarrow$
 $(250) + (0.5)$

$\downarrow$
 $0.5 \times 8 = 4.0(a^{-1})$

$8 \overline{) 250}$
 $8 \overline{) 31} - 2(a^1)$
 $3 - 7(a^0)$

$\Rightarrow (250.5)_{10} = (a^2 a^1 a^0 a^{-1})_8 = (372.4)_8$

(d) $(250.5)_{10}$ to $(\)_{16}$

$\downarrow$
 $0.5 \times 16 = 8.0(a^{-1})$

$16 \overline{) 250}$
 $15 - 10(a^1)$

$\Rightarrow (250.5)_{10} = (a^1 a^0 a^{-1})_{16} = (FA.8)_{16}$

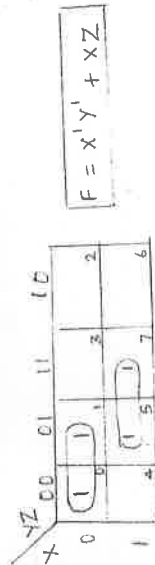
(6)

5) simplify the following expression using K-map (NOV'12)

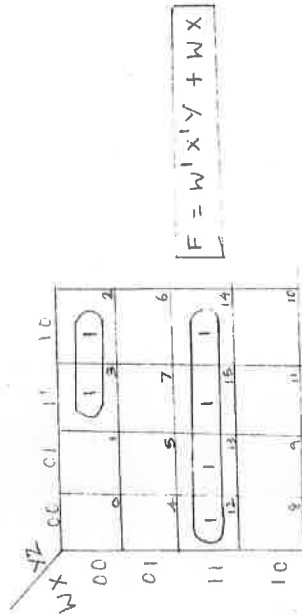
(a) $F(x, y, z) = \sum (1, 2, 3, 6, 7)$



(b) $F(x, y, z) = \sum (0, 1, 5, 7)$

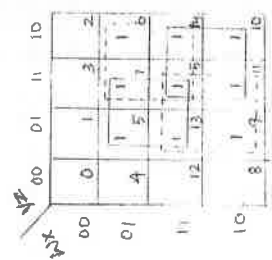


(c) $F(w, x, y, z) = \sum (2, 3, 12, 13, 14, 15)$



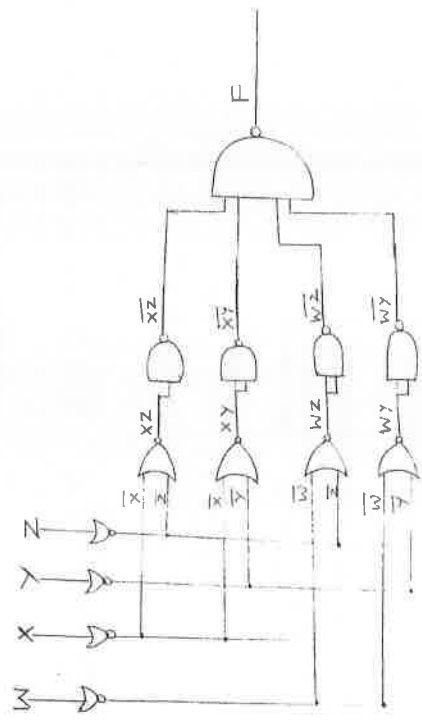
- 6) Reduce the following expression using k-map and implement them in Universal logic (8 marks)

$$\Sigma m(5, 6, 7, 9, 10, 11, 13, 14, 15)$$



$$F = XZ + XY + WZ + WY$$

Logic Implementation of $F = XZ + XY + WZ + WY$



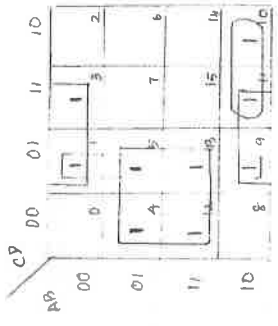
- 7) Simplify the following Boolean expression using k-map. (11 marks)

$$(a) WXY + YZ + XY'Z + X'Y$$



$$F = X'Y + YZ + XZ + WY$$

(b) $B'D + A'BC + AB'C + ABC'$



$$F = BC' + C'D + B'D + AB'C$$

12) Reduce the following Boolean expression. (APR'10)

$$(a) ABC + A'B'C + A'BC + ABC' + A'B'C' + A'B'C' \quad (5 \text{ MARKS})$$

$$\Rightarrow C(AB + A'B) + A'BC + C'(AB + A'B) \quad (\because x + x' = 1)$$

$$\Rightarrow C + A'BC + C' \quad (\because x + x' = 1)$$

$$\Rightarrow A'BC + 1$$

$$\Rightarrow 1 \quad (\because 1 + x = 1)$$

$$(b) BC + AC' + A'B + BCD \quad (4 \text{ MARKS})$$

$$\Rightarrow BC(1 + D) + AC' + A'B$$

$$\Rightarrow BC + AC' + A'B \quad (\because 1 + x = 1)$$

$$(c) (A + C + D)(A + C + D')(A + C' + D)(A + B')$$

$\Rightarrow$ Convert given expression to SOP

$$\Rightarrow A'C'D' + A'C'D + A'C'D' + A'B$$

$$\Rightarrow A'C'(D' + D) + A'C'D' + A'B$$

$$\Rightarrow A'C' + A'C'D' + A'B$$

$\Rightarrow$

13)

Theorem 1(b): $x \cdot x = x$

$$\begin{aligned} x \cdot x &= x \cdot x + 0 \quad (\because x + 0 = x) \\ &= x \cdot x + x \cdot x' \quad (\because x \cdot x' = 0) \\ &= x(x + x') \quad (\because x + x' = 1) \\ &= x \cdot 1 \\ &= x \end{aligned}$$

$$\boxed{x \cdot x = x}$$

Theorem 2(a): $x + 1 = 1$

$$\begin{aligned} x + 1 &= 1 \cdot (x + 1) \quad (\because 1 \oplus x = 1) \\ &= (x + x')(x + 1) \quad (\because x + x' = 1) \\ &= x + x' \cdot 1 \quad (\because x + 1 = 1) \\ &= x + x' \quad (\because x + x' = 1) \\ &= 1 \end{aligned}$$

Theorem 2(b): $x \cdot 0 = 0$ by duality

Theorem 3: $(x')' = x$

Boolean Laws

a) De Morgan's Theorem

(i) $\overline{AB} = \overline{A} + \overline{B}$ [The complement of a product is equal to the sum of the complements]

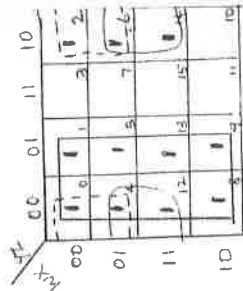
A	B	$\overline{AB}$	$\overline{A} + \overline{B}$
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0

9.

8) simplify the following Boolean function using K-map
(11) (NOV'11,

$$F(W, X, Y, Z) = \sum (1, 4, 6, 12, 8, 9, 14, 13, 95)$$

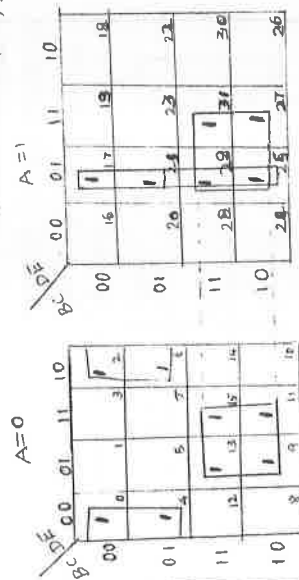
$$F(W, X, Y, Z) = \sum (0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$$



$$F = Y' + W'Z' + XZ'$$

9) simplify Boolean function: (11) (APR/MAY'11)

$$F(A, B, C, D, E) = \sum (0, 2, 4, 6, 9, 11, 13, 15, 17, 21, 25, 27, 29, 31)$$

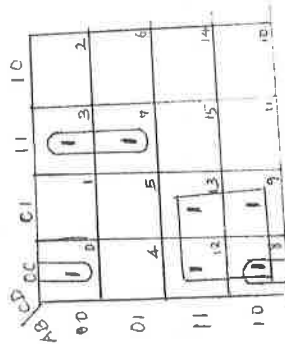


$$F = A'B'E' + A'BE + ABE + ADE$$

$$F = A'B'E' + BE + ADE$$

10

10) simplify the Boolean expression $F = \pi(0, 3, 7, 8, 9, 12, 13)$
(5) (APR'11) using Karnaugh map.



$$F = B'C'D + AC' + A'C'D$$

11) Simplify the Boolean Expression (6) (APR'11)

$$\bar{A}\bar{B}CD + ABCD + \bar{A}C\bar{D} + \bar{A}CD + \bar{A}\bar{B}D$$

$$\Rightarrow CD(AB + \bar{A}\bar{B}) + \bar{A}C(D + \bar{D}) + \bar{A}\bar{B}D$$

$$\Rightarrow CD + \bar{A}C + \bar{A}\bar{B}D$$

(14)

(ii) $\overline{A+B} = \overline{A} \cdot \overline{B}$

[The complement of a sum is equal to the product of the complements.]

A	B	$\overline{A+B}$	$\overline{A} \cdot \overline{B}$
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

(b) Commutative Law

Law 1: $A+B = B+A$ [This states that the order in which variables are ORed makes no difference in the output.]

A	B	$A+B$
0	0	0
0	1	1
1	0	1
1	1	1

B	A	$B+A$
0	0	0
0	1	1
1	0	1
1	1	1

(15)

Law 2: $A \cdot B = B \cdot A$ [The commutative law of multiplication states that the order in which the variables are ANDed makes no difference in the output.]

A	B	$A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

B	A	$B \cdot A$
0	0	0
0	1	0
1	0	0
1	1	1

(c) Associative Law

Law 1: $A+(B+C) = (A+B)+C$

This law states that in ORing of several variables, the result is the same regardless of the grouping of the variables.

A	B	C	$A+B$	$(A+B)+C$
0	0	0	0	0
0	0	1	0	1
0	1	0	1	1
0	1	1	1	1
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

A	B	C	$B+C$	$A+(B+C)$
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	1	1
1	0	0	0	1
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

Law 2: $(AB)C = A(BC)$

This law states that in ANDing several variables, the result is the same regardless of the grouping of variables.

A	B	C	AB	(AB)C
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	0	0
1	1	0	1	0
1	1	1	1	1

A	B	C	BC	A(BC)
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	0
1	0	0	0	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

(d) Distributive Law $[A(B+C) = AB+AC]$

This law states that ORing several variables and ANDing the result with a single variable is equivalent to ANDing the result with a single variable with each of the several variables and the ORing the product.

A	B	C	B+C	A(B+C)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

A	B	C	AB	AC	AB+AC
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	1	0	1
1	1	1	1	1	1

(17)

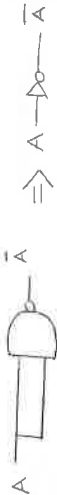
Show that both NAND gate and NOR gate are universal gates. (5 MARK)

(OR)
Realize the basic gates using NAND and NOR gates. (5) (NOV'11)

(OR)
Why NAND and NOR gate is said to be universal gates? Prove.

NAND is a universal gate, because any digital system can be implemented using it.

Implementation of NOT using NAND



Implementation of AND using NAND



Implementation of OR using NAND

